Raushan Kumar Singh

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Aratt Premier, Whitefield, Bangalore - 560048, Karnataka, India

Education

IIT (ISM) Dhanbad — Dhanbad, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.80 — *Jun 2020*

D.A.V. Public School — Patna, India
CBSE Higher Secondary Examination; Percentage: 93.2% — *2014 - 2016*

D.A.V. Public School — Patna, India
CBSE Secondary Examination; CGPA : 10 — *2013 - 2014*

Professional Experience

Jio Platforms Limited — Bangalore, India
Senior Software Development Engineer — *Aug 2020 - Present*

- **JioCinema (Jul 2022 - Present)**:
  - **Drove growth to 20M+ active subscribers** by leading the end-to-end implementation and rollout of a high-traffic subscription management system, load-tested for 10,000 req/s, ensuring 99.9% uptime.
  - **Reduced MTTR** to under 1 hour for SEV-1 and SEV-2 issues, managing live incidents on PagerDuty during peak traffic and delivering postmortem analysis to prevent future recurrence.
  - **Optimized system performance**, achieving 3x faster API response and 5x faster UI rendering, by implementing Redis-based locking, DB indexing, and batch processing to handle high concurrency and prevent RabbitMQ overload.
  - **Tech Stack:** NestJS, Typescript, RabbitMQ, Redis, Postgres, GCP, Kubernetes
- **JioWatch (Sep 2021 - Jul 2022)**:
  - **Achieved multi-tenancy** with streamlined tenant onboarding and data isolation using AsyncLocalStorage and Mongoose pre-hooks.
  - **Optimized livestream and VOD transcoding** by fine-tuning various FFMpeg parameters, enabling automatic GPU transcoding during high traffic (brought down latency to less than 7s, 10 times faster VOD transcoding using Tesla T4 GPU machines).
  - **Led the integration** of PassportJS to support multiple authentication methods, including Facebook, Google, email/password, and OTP-based login.
  - **Tech Stack:** NodeJS, Javascript, RabbitMQ, Redis, MongoDB, Nginx, ElasticSearch
- **JioMeet (Aug 2020 - Sep 2021)**:
  - **Implemented and deployed** SCI file upload server, eliminated JioCloud dependency, added support for upload of multiple file types.
  - **Enhanced chat functionality** by integrating Rocket.Chat for enterprise users (150+ enterprise customers)
  - **Achieved a significant reduction** of around 70% in monthly bugs through code refactoring, reducing cyclomatic complexity, and enhancing code coverage with unit tests.
  - **Tech Stack:** NodeJS, RabbitMQ, Redis, MongoDB, Docker, Chai, Sinon

Athabasca University — Edmonton, Canada
Research Intern (Machine Learning) -offered by MITACS — *May 2019 - Jul 2019*

- **Awarded CAD 8,000 research grant** for adapting and fine-tuning a pre-trained deep learning LSTM model to predict traffic flow, achieving an F1-score of 0.92, and conducting a simulation-based study to optimize traffic light control on main artery roads.
- Collaborated with Dr. Vivekanandan Suresh Kumar to analyze and optimize traffic flow patterns, leading to a **20% reduction in congestion** on main artery roads through data-driven traffic light control strategies.
- **Tech Stack:** Python, Keras, Numpy, Pandas, Jupyter Notebooks, Git

Skills

- **Languages/ Framework**: C++, C, Bash, MySQL, Shell scripting, Agile, JavaScript, TypeScript, NodeJS, NestJS, MongoDB, Postgres, ElasticSearch, Redis, RabbitMQ, Nginx, Akamai CDN
- **Tools and Libraries**: Git, Docker, Google cloud platform - Compute Engine, Cloud Storage, Pub/Sub, Artifact Registry, Kubernetes Engine, NPM, FFmpeg, Grafana, Loki, Promtail, TypeORM, Jaeger, OpenTelemetry
- **Massive Open Online Courses (MOOCs)**: Machine Learning by Stanford University (Grade 96.4%), Deep learning specialization by Andrew Ng(Grade 97%), Data Structures and Algorithms by San Diego University, GCP Big Data and ML Fundamentals(Grade 100%)
- **Soft Skills**: Leadership, Industrious, Versatile, Self-motivated, Creative, Problem Solving, Teamwork, Humour, Communication skills

Positions Of Responsibility & Achievements

- **Secretary** at ACM Students Chapter, *IIT(ISM) Dhanbad*
- **Technical Head** at Annual Cultural Fest, *Srijan'18*
- **Organizer** at Annual Techno-management Fest, *Concetto'18*